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# Deferred Transmission Investment Due to the Impact of Behind-the-Meter Distributed Generation and Energy Communities: Static and Multi-Period MILP-DC Formulation with CEI-Based Prioritization

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**Abstract:** Behind-the-meter (BTM) generation and energy communities reduce the demand that transmission networks must carry, yet expansion planning still treats that demand as given. This paper proposes a planning methodology that represents BTM self-supply as a virtual transmission resource, introduced in the expansion problem as a nodal reduction of net demand, and values the investment it allows planners to avoid or postpone. The methodology combines static and multi-period expansion models, formulated as mixed-integer linear programs with a DC network representation, with a Community Energy Index (CEI) that ranks nodes by solar resource, supply deficit and loading of adjacent corridors to indicate where BTM deployment should be promoted. It applies to any network with significant BTM penetration and is illustrated on two systems: the six-bus Garver test system, which isolates the mechanism, and the Colombian 500 kV backbone, a system where new large loads, including data centres, are expected to connect behind the meter rather than directly to the grid. The results show that this deferred investment credit grows in steps rather than in proportion to penetration, depends on where the resource is located, and combines avoided and postponed investment, with a floor set by the connection of new load. The approach offers planners a way to align energy community incentives with transmission expansion.

**Keywords:** transmission expansion planning; behind-the-meter generation; energy communities; deferred investment credit; virtual transmission resource; mixed-integer linear programming; community energy index

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## 1. Introduction

Transmission expansion planning (TEP) decides which circuits to build, and when, so that the network can carry the demand projected over a long horizon [1,2]. That demand enters the problem as a fixed input. The premise is losing ground as behind-the-meter (BTM) generation and storage spread: when users cover part of their consumption locally, the demand the high-voltage grid has to serve is smaller than the gross projection, and if the reduction happens at the right nodes it can postpone a corridor or remove it from the plan. This paper calls the resulting saving the deferred transmission investment credit, and proposes a methodology to quantify it and to decide where BTM deployment is worth promoting.

The TEP literature has refined the network model and the solution methods, from the transportation and hybrid formulations to the full DC model with disjunctive constraints [2–4], bi-level formulations for market environments [5], and reliability-constrained variants [6,7]. More recent work adds other assets to the expansion decision, such as line switching [8], FACTS devices [9] and transmission-level storage [10]. In all of them demand is

exogenous, and any resource other than the line is a supply-side or control element operated by the planner.

The idea that dispersed resources can defer network investment is older than most of this work. Lee et al. [11] examined how dispersed storage and generation change transmission requirements, Ma et al. [12] did the same for distribution planning, and Shugar [13] evaluated the distributed benefits of photovoltaic plants connected to a distribution feeder. The concept was later formalized in several directions. Li and Zielke [14] valued small load reductions through the one-year deferral method; De Oliveira-De Jesus et al. proposed a general framework for remunerating the costs and benefits of distributed generation [15], a long-term cost allocation method [16] and a fuzzy multicriteria planning approach [17], unified in [18]; Khodr et al. [19] combined Benders decomposition and fuzzy multicriteria analysis for the same purpose, and Mendez et al. [20] assessed deferral under uncertainty through Monte Carlo simulation. A parallel line values the contribution of renewables and storage to generation adequacy, the capacity credit [21,22], which Liu et al. [23] extended to multi-area systems where transfer limits make the credit location-dependent.

All of these works credit a resource for what it injects. Whether it is a distributed generator, a central renewable plant or a storage unit, it is connected to the grid, it appears in the nodal balance as an injection, and its benefit is measured against that injection. BTM self-supply does not fit that description: it never reaches the network, it is not dispatched by the operator, and its only trace in the planning problem is a smaller withdrawal at the node. The energy community literature does treat resources behind the meter, through multicriteria design and operation [24–26] or the dispatch of a community that owns its own closed distribution system [27], but it stays at the community scale and does not measure an effect on high-voltage expansion. Table 1 places these contributions side by side.

**Table 1.** State of the art. Level: Tx transmission, Dx distribution, Gen generation adequacy, EC energy community. Credit: network, avoided or deferred network investment; capacity, contribution to adequacy

| Reference | Year | Level | Resource | Credit | Multi-period | Siting priority |
|---|---|---|---|---|---|---|
| [1] | 1970 | Tx | none | – | no | no |
| [3] | 2003 | Tx | none | – | no | no |
| [4] | 2010 | Tx | none | – | yes | no |
| [8] | 2020 | Tx | line switching | – | no | no |
| [10] | 2023 | Tx | storage | – | yes | no |
| [9] | 2024 | Tx | FACTS | – | no | no |
| [11] | 1979 | Tx | dispersed generation | network | no | no |
| [12] | 1979 | Dx | dispersed generation | network | no | no |
| [13] | 1990 | Dx | grid-connected PV | network | no | no |
| [20] | 2003 | Dx | DG | network | no | no |
| [14] | 2005 | Tx, Dx | load reduction | network | no | no |
| [16] | 2005 | Dx | DG | network | no | no |
| [15] | 2006 | Dx | DG | network | no | no |
| [17] | 2006 | Dx | DG | network | no | no |
| [19] | 2009 | Dx | DG | network | no | no |
| [21] | 1999 | Gen | wind | capacity | no | no |
| [23] | 2021 | Gen | RES and storage | capacity | no | partial |
| [24] | 2022 | EC | community DER | – | yes | no |
| [27] | 2024 | EC | BTM, closed system | – | no | no |
| This paper | 2026 | Tx | BTM self-supply | network | yes | yes |

The table makes the gap explicit. Network credits have been computed for resources that inject, almost always at the distribution level; the works that model resources behind the meter do not reach the transmission network; and none of them ranks locations,

although the value of a demand reduction clearly depends on how close it is to a congested corridor. No contribution represents BTM self-supply as a reduction of net demand inside a transmission expansion model and values the investment it avoids or postpones.

The methodology is relevant wherever BTM penetration is large enough to change transmission requirements, which is also where a competing effect has to be considered. New large loads, data centres in particular, could offset the demand reduction: global data centre consumption was projected to grow from about 460 TWh in 2022 to between 620 and 1050 TWh in 2026 [28,29]. Whether that happens depends on how those loads connect. In countries such as Colombia, where the regional data centre market is still emerging [30], connection capacity at the main load centres is scarce and the solar resource is good, a new large load faces long waits and high costs to connect directly to the transmission system, and the rational choice is to supply itself behind the meter. This paper takes that as an explicit premise: in such systems, data centres and similar loads connect behind the meter and add to the BTM resource rather than offsetting it.

To fill the gap, this paper makes four contributions. First, a static mixed-integer linear expansion model with a DC network representation (MILP-DC) in which nodal demand enters as $d_i(1-\alpha_i)$, so that BTM penetration becomes an explicit planning input and the avoided investment is read directly from the optimal plan. Second, a multi-period extension with per-period construction decisions and a present-value objective, which separates investment that is avoided from investment that is only postponed. Third, a Community Energy Index (CEI) that combines solar potential, nodal supply deficit and the loading of adjacent corridors, and guides deployment towards the nodes where the credit per unit of penetration is largest. Fourth, the application of the methodology to two cases: the six-bus Garver system, where the mechanism can be followed circuit by circuit, and the 500 kV backbone of the Colombian system, as an example of a real network with significant BTM potential.

The rest of the paper is organized as follows. Section 2 presents the conceptual framework, Section 3 the models and the index, Section 4 the two case studies, Section 5 the results, and Section 6 the conclusions.

## 2. Conceptual Framework

### *2.1. The behind-the-meter resource as a virtual transmission asset*

A transmission corridor is built to carry the difference between what is generated and what is consumed on each side of it. Any resource that reduces consumption at the importing end therefore has the same effect on the corridor as an increase in its capacity, without a single kilometre of conductor being installed. This is the sense in which BTM self-supply is treated here as a virtual transmission asset: it does not inject power into the backbone, and it is not dispatched by the system operator, but it changes the quantity the backbone has to transport (Figure 1).

Formally, let $d_i$ be the gross demand projected at node $i$ and $\alpha_i \in [0,1]$ the fraction of that demand served locally by generation and storage operated behind the meter. The demand seen by the transmission system is

$$d_i^{\text{net}} = d_i\,(1-\alpha_i) \tag{1}$$

The parameter $\alpha_i$ aggregates every form of self-supply behind the point of connection: individual self-generation of any size, collective schemes in which several users share a single metering boundary, and energy communities in general. From the point of view of the backbone the legal or commercial arrangement is irrelevant, because all of them produce the same effect, a smaller net withdrawal at the node. The same holds for a large load that connects with its own generation behind the meter: its consumption is served locally and does not reach the transmission network.

Two features of this representation matter for what follows. First, $\alpha_i$ is an exogenous planning input, not a decision variable: the planner does not choose how much self-generation is installed, but can ask what an expansion plan would look like under a given

penetration. Second, the effect is local. A reduction at an exporting node with spare capacity changes nothing, while the same reduction at an importing node fed by a loaded corridor can remove a circuit from the optimal plan. This asymmetry is what motivates a prioritization index.

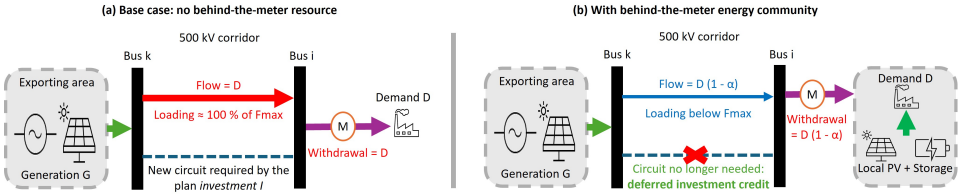



**Figure 1.** Behind-the-meter self-supply as a virtual transmission asset: local generation reduces the net demand carried by the backbone and relieves the adjacent corridors

### *2.2. Avoided investment and postponed investment*

The benefit that a given penetration produces is not the same quantity in a single-horizon study and in a study that follows the system year by year, and the two are often confused. A static model compares the optimal plan for a target year with and without BTM resources. The difference is investment that the plan no longer contains, that is, avoided investment:

$$C^{\mathrm{av}}(\alpha) = I^{\mathrm{base}} - I(\alpha) \tag{2}$$

where $I^{\mathrm{base}}$ is the cost of the optimal plan with $\alpha_i = 0$ for all nodes and $I(\alpha)$ the cost of the optimal plan under the penetration vector $\alpha$.

A multi-period model decides not only which circuits to build but in which year. Under a penetration that grows over time, a circuit that the base plan requires early may be pushed towards the end of the horizon, or out of it. Because the planner discounts, a circuit built later is worth less today, and the benefit is measured as a difference of present values:

$$C^{\mathrm{def}}(\alpha) = \sum_{t \in T} \frac{I_t^{\mathrm{base}}}{(1+r)^{\,t-1}} - \sum_{t \in T} \frac{I_t(\alpha)}{(1+r)^{\,t-1}} \tag{3}$$

with $r$ the discount rate and $I_t$ the investment committed in period $t$. Equation (3) captures both effects at once: a circuit that disappears from the plan contributes its full cost, and a circuit that is merely delayed contributes the financial value of the delay, in the spirit of the one-year deferral method [14]. Throughout the paper, deferred credit refers to (3) and avoided investment to (2).

### *2.3. Where to deploy: uniform and prioritized allocation*

Since the effect of a reduction depends on the node, a total penetration budget can be distributed in different ways. Two allocations are compared. The uniform allocation assigns the same fraction everywhere, $\alpha_i = \bar{\alpha}$, and represents a policy that is blind to the state of the network. The prioritized allocation distributes the same aggregate penetration in proportion to a composite index,

$$\alpha_i = \min\left(\bar{\alpha}\,\frac{CEI_i}{\overline{CEI}},\ \alpha^{\max}\right) \tag{4}$$

where $CEI_i$ is the Community Energy Index of node $i$, $\overline{CEI}$ its mean over the nodes with demand, and $\alpha^{\max}$ a cap that prevents any single node from being assigned an implausible penetration. The index combines what makes a node attractive for a community and what makes it valuable for the grid: the solar resource available at its location, the gap between local generation and local demand, and how loaded the corridors around it already are. Its formal definition is given in Section 3. Comparing the two allocations at

equal aggregate penetration isolates the value of siting, separately from the value of the resource itself.

### 2.4. Method in two stages

The method has two stages, shown in Figure 2. In the first stage the state of the network in the target year, together with the solar resource, produces the index and the two penetration vectors. In the second stage the expansion models are solved, first with no BTM resources to obtain the base plan and then for each penetration level and each allocation strategy. The static model returns the avoided investment and the multi-period model returns the construction schedule and the deferred credit. Nothing in the second stage feeds back into the first: the index is computed from the base case, so the prioritization does not depend on the plan it helps to modify.

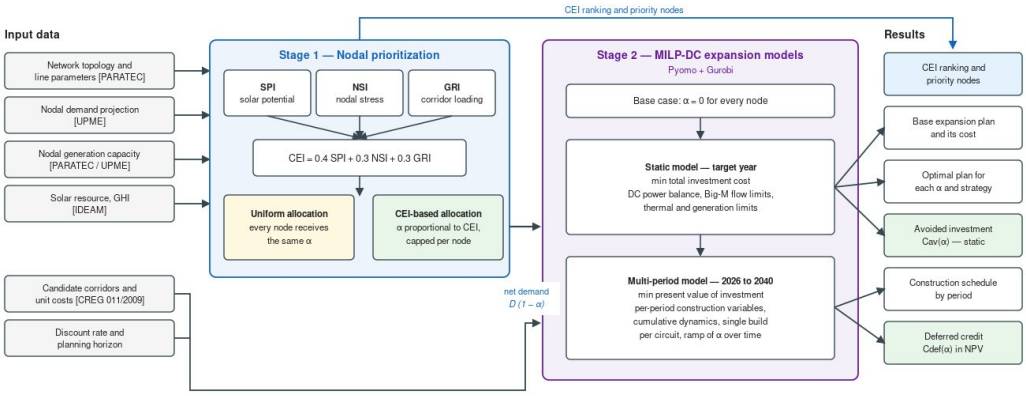



**Figure 2.** General framework: construction of the Community Energy Index and evaluation of the static and multi-period expansion plans under each penetration level

## 3. Methodology

This section presents the two expansion models and the prioritization index. The notation is summarized in Appendix A.

### 3.1. Assumptions

The models share a set of assumptions that define the scope of the analysis. The network is represented by a lossless DC power flow, which is standard in expansion planning because the decision of interest is the discrete choice of circuits rather than the operating point. Only the 500 kV level is modelled, so reinforcements at lower voltages and transformation capacity are outside the scope. Circuits are limited by their thermal rating and no security criterion is imposed, which means the plans reported here are the minimum set of circuits needed under normal operation. Generation is bounded by the capacity available at each node and is dispatched without cost, so the optimizer selects the dispatch that best relieves the network among all feasible ones. Finally, the penetration $\alpha$ is an exogenous planning input rather than a decision variable: the models answer what the network would require under a given level of behind-the-meter self-supply, not how much self-supply a community would choose to install.

### 3.2. Static MILP-DC model

Let $\mathcal{I}$ be the set of nodes, $\mathcal{J}$ the set of corridors, and $\mathcal{K} = \{1, \ldots, K\}$ the set of circuit positions available in each corridor, of which the first $n_j^0$ are already in service. The binary variable $y_{j,k}$ takes the value one when circuit $k$ of corridor $j$ is in service, $f_{j,k}$ is the power flow through it, $g_i$ is the power injected at node $i$, and $\theta_i$ is the voltage angle. The problem minimizes the cost of the circuits that have to be built:

$$\min \sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{K},\, k > n_j^0} c_j \, y_{j,k} \tag{5}$$

subject to the nodal power balance, written with the net demand of Equation (1),

$$\sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{K}} S_{ij} \, f_{j,k} \; + \; g_i \; = \; d_i \, (1 - \alpha_i), \qquad \forall i \in \mathcal{I} \tag{6}$$

where $S_{ij}$ is the node-corridor incidence, equal to $+1$ if $i$ is the sending end of $j$, $-1$ if it is the receiving end, and zero otherwise. The DC flow through a circuit is imposed only when that circuit exists, which is written with a disjunctive Big-M formulation:

$$-M_j \, (1 - y_{j,k}) \; \le \; f_{j,k} - B_j \, (\theta_a - \theta_b) \; \le \; M_j \, (1 - y_{j,k}), \qquad \forall j \in \mathcal{J}, \; \forall k \in \mathcal{K} \tag{7}$$

with $a$ and $b$ the ends of corridor $j$ and $B_j$ its susceptance. When $y_{j,k} = 1$ the constraint reduces to the DC flow equation, and when $y_{j,k} = 0$ it becomes redundant while the thermal limit forces the flow to zero:

$$-F_j^{\max} \, y_{j,k} \; \le \; f_{j,k} \; \le \; F_j^{\max} \, y_{j,k}, \qquad \forall j \in \mathcal{J}, \; \forall k \in \mathcal{K} \tag{8}$$

The remaining constraints bound generation, fix the circuits already in service, set the angular reference at the node with the largest net export, and cap the total investment when a budget applies:

$$0 \; \le \; g_i \; \le \; G_i^{\max}, \qquad \forall i \in \mathcal{I} \tag{9}$$

$$y_{j,k} = 1, \qquad \forall j \in \mathcal{J}, \; \forall k \le n_j^0 \tag{10}$$

$$\theta_{\text{ref}} = 0 \tag{11}$$

$$\sum_{j \in \mathcal{J}} \sum_{k > n_j^0} c_j \, y_{j,k} \; \le \; \Gamma \tag{12}$$

The parameter $M_j$ deserves a comment, because a value that is too small silently removes feasible plans. Since the left-hand side of (7) is bounded by $B_j \, |\theta_a - \theta_b|$, it is enough to take $M_j = B_j \, \Delta\theta^{\max}$, where $\Delta\theta^{\max}$ bounds the angular difference between any pair of nodes. That bound is computed from the existing network as the largest shortest path between two nodes, weighting each corridor by the angular opening it develops at full load, $X_j F_j^{\max}$, plus twice the opening of the largest candidate corridor to account for the substations that are only reachable through a new circuit.

### *3.3. Multi-period MILP-DC model*

The static model answers whether a circuit is needed at a target year. The multi-period model asks when. The horizon is divided into periods $t \in \mathcal{T} = \{1, \ldots, T\}$, one per year. The binary variable $y_{j,k,t}$ indicates that circuit $k$ of corridor $j$ is built in period $t$, and $x_{j,k,t}$ indicates that it is available in that period. The objective is the present value of the investment:

$$\min \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}} \sum_{k > n_j^0} \frac{c_j \, y_{j,k,t}}{(1 + r)^{\, t-1}} \tag{13}$$

Equations (6) to (11) are replicated for every period, with $x_{j,k,t}$ in place of $y_{j,k}$ and with demand and generation evaluated at each year. Two constraints link the periods. The first

states that a circuit available in $t$ is one that was already available in $t-1$ or was built in $t$, starting from the circuits already in service:

$$x_{j,k,t} = x_{j,k,t-1} + y_{j,k,t}, \qquad x_{j,k,0} = \begin{cases} 1 & k \le n_j^0 \\ 0 & k > n_j^0 \end{cases} \tag{14}$$

The second prevents a circuit from being built more than once, which also keeps $x_{j,k,t}$ binary without declaring it as such:

$$\sum_{t \in \mathcal{T}} y_{j,k,t} \;\le\; 1, \qquad \forall j \in \mathcal{J},\ \forall k \in \mathcal{K} \tag{15}$$

Demand and generation are interpolated linearly between the anchor years of the projection, and behind-the-meter penetration is introduced as a ramp that starts at zero in the first period and reaches its final value at the end of the horizon, which represents progressive adoption rather than an installed base:

$$\alpha_{i,t} = \alpha_i \, \frac{t-1}{T-1}, \qquad \forall i \in \mathcal{I},\ \forall t \in \mathcal{T} \tag{16}$$

The deferred credit of Equation (3) is obtained by solving (13) with $\alpha_i = 0$ and then with the penetration vector under evaluation, and comparing the two present values.

### *3.4. Community Energy Index*

The index ranks nodes by how much a reduction of net demand at that node is worth to the transmission system, and by how easily a community could produce that reduction. It combines three normalized sub-indices.

The solar potential index rewards the availability of the resource, using the global horizontal irradiance at the node:

$$SPI_i = \frac{GHI_i}{\max_{n \in \mathcal{I}} GHI_n} \tag{17}$$

The nodal stress index measures how much of the local demand is not covered by local generation, so that exporting nodes score zero:

$$NSI_i = \frac{1}{\eta_N} \max\!\left(0,\ \frac{d_i - G_i^{\max}}{d_i}\right), \qquad \eta_N = \max_{n \in \mathcal{I}} \max\!\left(0,\ \frac{d_n - G_n^{\max}}{d_n}\right) \tag{18}$$

The grid reinforcement index accumulates the loading of the corridors incident to the node, capped at full load so that a single saturated corridor does not dominate the sum:

$$GRI_i = \frac{1}{\eta_G} \sum_{j \in \Omega_i} \min\!\left(\frac{|\hat{f}_j|}{\max(n_j^0, 1)\, F_j^{\max}},\ 1\right) \tag{19}$$

where $\Omega_i$ is the set of corridors incident to node $i$, $\hat{f}_j$ the flow of the base case and $\eta_G$ the largest value of the sum over the nodes; the maximum with one prevents a zero capacity for candidate corridors with no circuit in service. The three components are combined with fixed weights:

$$CEI_i = 0.4\, SPI_i + 0.3\, NSI_i + 0.3\, GRI_i \tag{20}$$

The weights give priority to the resource, because a community cannot be built where there is no resource to harvest, and split the remainder evenly between the two grid-related terms, which measure the same congestion from the node side and from the corridor side. This weighting follows the practice of multicriteria studies on energy systems [25,26]. The index feeds the allocation rule of Equation (4), and nodes without demand receive no penetration. The cap is set at $\alpha^{max} = 0.40$ for the Colombian backbone; it is not applied to the Garver system, for the reason given in Section 4.1.

### 3.5. Procedure

The complete procedure is given in Algorithm 1.

**Algorithm 1** Deferred transmission investment credit with CEI-based prioritization

**Require:** Network topology and line parameters; nodal demand and generation projections; candidate corridors and unit costs; global horizontal irradiance; discount rate $r$ and horizon $T$

**Ensure:** Base plan, avoided investment $C^{\mathrm{av}}(\alpha)$, construction schedule and deferred credit $C^{\mathrm{def}}(\alpha)$

1: Assemble the input data and compute $B_j$, $F_j^{\max}$, $c_j$ and $M_j$ for every corridor
2: Solve the static model (5)–(12) with $\alpha_i = 0$ for the target year
3: Store the base plan, its cost $I^{\mathrm{base}}$ and the corridor loadings $\hat{f}_j$
4: Compute $SPI_i$, $NSI_i$ and $GRI_i$ from the base case using (17)–(19)
5: Obtain $CEI_i$ from (20) for every node with demand
6: **for** each aggregate penetration level $\bar{\alpha}$ **do**
7: Build the uniform vector $\alpha_i = \bar{\alpha}$ and the CEI-based vector from (4)
8: **for** each allocation strategy **do**
9: Solve the static model with the net demand $d_i(1 - \alpha_i)$
10: Compute the avoided investment $C^{\mathrm{av}}(\alpha)$ from (2)
11: Solve the multi-period model (13)–(16) with the penetration ramp
12: Compute the deferred credit $C^{\mathrm{def}}(\alpha)$ from (3)
13: Compare both strategies at equal aggregate penetration and identify the corridors that leave the plan at each level

Steps 2 and 10 answer how much investment a given penetration avoids, steps 11 and 12 separate what is avoided from what is only postponed, and step 13 isolates the value of siting the resource.

## 4. Case Study

The models are applied to two systems: the six-bus Garver system, where the mechanism can be observed in isolation, and an equivalent of the 500 kV backbone of the Colombian National Interconnected System.

### 4.1. Garver test system

The six-bus system introduced by Garver [1] is the standard benchmark for expansion planning. It has 760 MW of demand and 760 MW of available capacity, distributed as shown in Table 2, and 15 possible corridors, six of which already carry a circuit. Each corridor admits up to five circuits, and the corridor data are given in Appendix B. Bus 6 is isolated in the initial configuration and holds most of the generation, so the plan must connect it: generation far from the load and a network that has to be built to reach it, which is the situation the methodology is meant to address.

**Table 2.** Demand and generation capacity of the Garver system

| Bus | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Demand (MW) | 80 | 240 | 40 | 160 | 240 | 0 |
| Generation capacity (MW) | 50 | 0 | 165 | 0 | 0 | 545 |

The static model is solved for penetrations from 0 to 50 % in steps of 10 %, the range over which the plan of this system changes. For the multi-period model, demand grows linearly from 55 % of the values in Table 2 to their full value over a ten-year horizon, with a discount rate of 10 %. In this system the CEI allocation is applied without the cap $\alpha^{max}$. Buses 2, 4 and 5 have almost the same index, so from an aggregate penetration of about

34 % the cap would bind at all three importing buses, and at 50 % also at bus 1, and the CEI vector would collapse into a nearly flat 40 %, no longer representing the aggregate level being evaluated. In the Colombian network the cap binds only at the highest level studied and at a few nodes, so the allocation keeps its spatial differentiation and the cap is retained.

### *4.2. The Colombian 500 kV backbone*

Colombia is a useful example because it combines the two conditions the methodology requires. Its backbone links the hydro resources of the Andean region and the wind and solar potential of the Caribbean and La Guajira to load centres several hundred kilometres away, and the reference expansion plan foresees reinforcements on several of those corridors [31]. At the same time, in about a decade the country has built a regulatory framework that turns self-supply into a recognized system activity. Resolution CREG 030 of 2018 [32] first regulated small-scale self-generation; Resolution CREG 174 of 2021 [33] consolidated the rules for small-scale (AGPE, below 1 MW) and large-scale (AGGE) self-generation; Law 2099 of 2021 [34] declared non-conventional renewable sources of public utility; Decree 2236 of 2023 [35] and Resolution MME 40509 of 2024 [36] set the legal basis and the registry of energy communities; Resolution CREG 101 072 of 2025 [37] introduced the collective self-generator and virtual boundary aggregation, with a target of 1 GW from energy communities; and Resolution MME 40178 of 2026 [38] recognizes the geographic concentration of the generation matrix as a systemic risk. In this case study, $\alpha_i$ represents the joint effect of AGPE, AGGE and collective self-generation.

The premise stated in Section 1 is applied explicitly: the demand projection contains no large load connected directly to the backbone. Data centres and similar loads that enter the system over the horizon are assumed to connect behind the meter with their own supply, so they do not add to the nodal demand $d_i$ that the network has to serve.

The network is represented by the 21 substations that operate at 500 kV and the 4 substations associated with projects in the reference expansion plan, for a total of 25 nodes. Circuits connecting the same pair of substations are grouped into a corridor, whose susceptance and rating are the average of its circuits; the differences between parallel circuits are below 5 % in all cases. This yields 27 existing corridors carrying 31 circuits, and 10 candidate corridors that correspond to planned projects. Line parameters and ratings are taken from the public technical database of the system operator [39]: ratings range from 1650 to 2165 MVA and lengths from 36 to 275 km. As in the Garver case, each corridor admits up to five circuits, so reinforcing an existing corridor is also an option. The corridor data are given in Appendix B.

Investment costs follow the reference construction units of the regulator [40], expressed in December 2008 US dollars. The unit cost of a 500 kV single-circuit line is 259,944 USD/km at the lowest terrain level and 283,990 USD/km at the intermediate level, which gives costs per circuit between 10.2 and 78.1 MUSD depending on the length of the corridor. The figures reported in Section 5 are therefore not current dollars; since every corridor is costed with the same unit prices, indexing them would scale the whole plan and every saving by the same factor, leaving the comparison between strategies and the relative magnitude of the deferred credit unchanged. The budget constraint (12) is set well above the optimal plan and is never active.

### *4.3. Nodal demand and generation*

Demand is projected by the planning unit at the area level [41], and generation capacity is known plant by plant, so both have to be distributed over the nodes. The allocation used here is an approximation, and it is the input that most conditions the results, so the criteria are stated explicitly.

Demand is distributed within each area in proportion to the load each substation actually serves, that is, its transformation capacity towards the lower voltage levels and the urban centre it supplies. Substations that are essentially transit or generation nodes receive little or no demand. Generation is assigned to the substation where each plant connects,

and the resulting vector is rescaled so that the national total matches the projection for each anchor year. This is the main difference with respect to an allocation proportional to the connected transmission capacity, which distributes residual generation over transit substations and produces load centres that appear to be self-sufficient. Table 3 shows the resulting allocation.

**Table 3.** Nodal allocation of demand and generation capacity. Capacity is the installed value assigned to each substation before rescaling to the national total; the 1050 MW of Colectora and a further 1200 MW at Antioquia (completion of Ituango) are added in 2034

| Node | Substation | Area | Area demand (%) | Capacity (MW) |
|---|---|---|---|---|
| N3 | Bacatá | Oriental | 45 | 1000 |
| N13 | Nueva Esperanza | Oriental | 45 | 1213 |
| N16 | Primavera | Oriental | 10 | 600 |
| N23 | Norte | Oriental | 0 | 0 |
| N11 | Medellín | Antioquia | 45 | 120 |
| N8 | Heliconia | Antioquia | 30 | 80 |
| N5 | Cerromatoso | Antioquia | 10 | 340 |
| N2 | Antioquia | Antioquia | 10 | 1200 |
| N15 | Porce III | Antioquia | 5 | 1100 |
| N17 | Sabanalarga | Caribe | 30 | 1400 |
| N4 | Bolívar | Caribe | 24 | 900 |
| N7 | El Copey | Caribe | 16 | 400 |
| N12 | Nueva Cuestecitas | Caribe | 10 | 400 |
| N6 | Chinú | Caribe | 8 | 300 |
| N18 | Sahagún | Caribe | 7 | 0 |
| N9 | La Loma | Caribe | 5 | 300 |
| N24 | Carreto | Caribe | 0 | 0 |
| N25 | Nueva Magangué | Caribe | 0 | 0 |
| N22 | Colectora | Caribe | 0 | 1050 |
| N21 | Sogamoso | Nordeste | 65 | 1000 |
| N14 | Ocaña | Nordeste | 35 | 100 |
| N20 | San Marcos | Suroccidente | 33 | 300 |
| N1 | Alférez | Suroccidente | 32 | 400 |
| N10 | La Virginia | Suroccidente | 30 | 600 |
| N19 | San Carlos | Suroccidente | 5 | 2200 |

Three points follow from this allocation. The generation of the Guavio and Chivor complexes reaches the Bogotá area through the 230 kV ring, so it is assigned to the two 500 kV substations of that area, while the Norte substation is left with no generation because the project that would connect it is precisely one of the candidates. The Ituango complex and the Colectora wind hub enter in 2034, which is when their capacity is added at the Antioquia and Colectora nodes. And the conditioned demand of the Bogotá savanna, which today cannot be connected because the associated works are not in service, is represented at the Norte node, growing linearly to 400 MW by 2034 and following the scenario ramp thereafter.

Solar irradiance is taken from the national solar atlas [42] and assigned by area, between 4.3 and 4.8 kWh/m$^2$ per day in the Andean areas and 5.5 kWh/m$^2$ per day in the Caribbean, with 6.2 kWh/m$^2$ per day for the two substations located in La Guajira.

### *4.4. High-electrification scenario*

The reference projection alone does not stress the backbone enough to make the question meaningful: under the high-growth trajectory of the planning unit, the 2040 plan contains only the 24.7 MUSD circuit that connects the Norte substation, and no transfer

corridor needs reinforcement, although the most loaded one reaches 93 % of its rating. The deferred credit is only meaningful where expansion is needed, so the models are applied to a high-electrification scenario built on top of the reference projection. Demand grows faster in the areas where electrification and urban growth concentrate, and new capacity is added where the resource is, that is, in the Caribbean and La Guajira for wind and solar and, to a lesser extent, in the south-west. The growth factors are applied as a ramp, so the initial year keeps the projected values and the factor is reached in full at the end of the horizon. Table 4 reports them.

**Table 4.** Growth factors of the high-electrification scenario and resulting balance in 2040

| Area | D factor | G factor | Demand 2040 (MW) | Capacity 2040 (MW) |
|---|---|---|---|---|
| Oriental | 1.8 | 1.0 | 7571 | 4084 |
| Antioquia | 1.5 | 1.2 | 3683 | 7037 |
| Caribe | 1.4 | 2.0 | 8873 | 13790 |
| Nordeste | 1.4 | 1.2 | 3051 | 1916 |
| Suroccidente | 1.4 | 1.3 | 4746 | 6605 |
| Total | | | 27925 | 33433 |

The logic of the scenario is transparent: demand grows at the centre and capacity grows at the periphery, so the quantity to be transported grows faster than either of them. The system moves from 12,234 MW of demand and 21,271 MW of capacity in 2026 to 27,925 MW and 33,433 MW in 2040. The Oriental area, which concentrates the Bogotá load, ends the horizon with a deficit of 3487 MW that has to be imported through the backbone, while the Caribbean ends it with a surplus of 4917 MW.

The multi-period horizon covers 2026 to 2040, one period per year, with a discount rate of 10 %. Demand and generation between the anchor years of the projection are interpolated linearly. The angular reference is placed at San Carlos, the node with the largest net export at the target year.

The static model is solved for penetrations from 0 to 30 % in steps of 5 %; beyond 30 % only the connection of new load remains in the plan.

### *4.5. Implementation*

Both models were written in Pyomo [43] and solved with Gurobi [44], with an optimality gap of 0.3 % for the static model and 0.5 % for the multi-period one. The static instance has 185 binary variables, while the multi-period instance has 2,775 binary variables, plus 2,775 availability variables that remain binary by construction. Both instances are solved within 8 seconds on a computer equipped with an AMD Ryzen 3 processor and 8 GB of RAM. In the case studies, the multi-period model (step 11 of Algorithm 1) was solved for final penetrations of 10 and 20 % under the uniform allocation.

## 5. Results and Discussion

### *5.1. Garver test system*

Applying the static model to the Garver system with no behind-the-meter resource reproduces the canonical solution of the literature: four circuits on corridor 2–6, one on 3–5 and two on 4–6, seven circuits in total for 1000 kUSD. The plan coincides, circuit by circuit, with the one reported by Garver [1] and later replicated by Villasana et al. [2], Alguacil et al. [3] and Vargas-Robayo et al. [7], which verifies the formulation and its implementation before moving to a larger network.

The interest of the test system, however, is that the deferral mechanism can be inspected by hand. Table 5 reports the optimal investment as a function of penetration under the two allocation strategies.

**Table 5.** Static results on the Garver system by penetration level and allocation strategy

| $\alpha$ | $d^{\text{net}}$ (MW) | Uni. (kUSD) | CEI (kUSD) | Uni. saving (kUSD) | CEI saving (kUSD) |
|---|---|---|---|---|---|
| 0 % | 760 | 1000 | 1000 | 0 | 0 |
| 10 % | 684 | 850 | 850 | 150 | 150 |
| 20 % | 608 | 700 | 700 | 300 | 300 |
| 30 % | 532 | 700 | 550 | 300 | 450 |
| 40 % | 456 | 450 | 450 | 550 | 550 |
| 50 % | 380 | 300 | 300 | 700 | 700 |

Figure 3 shows the plan behind each row of Table 5 under the uniform allocation, and it reveals three features that reappear later on the real network. The first is that the cost falls in steps, because what leaves the plan is a whole circuit: at 10 % one of the four circuits of corridor 2–6 is no longer needed (panel c). The second is that the composition of the plan does not shrink monotonically even though its cost does. At 20 % the two circuits of corridor 4–6 disappear, but corridor 2–6 returns to four circuits and carries the output of bus 6 on its own (panel d); at 30 % one circuit of 4–6 comes back and 2–6 drops to three, a different plan with the same cost of 700 kUSD (panel e); and at 40 % the model abandons corridors 2–6 and 3–5 altogether and connects the remote generator through three circuits on 4–6 (panel f). Each penetration level therefore produces its own optimal topology, not a trimmed version of the previous one. The third feature is the value of siting: at 30 % the uniform allocation is still short of removing another circuit, while the CEI allocation, by concentrating the same aggregate penetration on the importing buses 2, 5 and 4, removes one more and reaches 550 kUSD (Figure 4, panel e). At 40 % and 50 % several plans share the optimal cost; the figures show the one returned by the solver.

Over a ten-year horizon with the same 10 % discount rate, the multi-period model on the same system produces a base plan with a present value of 823 kUSD, which falls to 730 kUSD at a final penetration of 10 % and to 643 kUSD at 20 %, that is, deferred credits of 93 and 180 kUSD. The nominal investment falls in steps while the present value falls continuously, because circuits that are still built are pushed towards later periods.

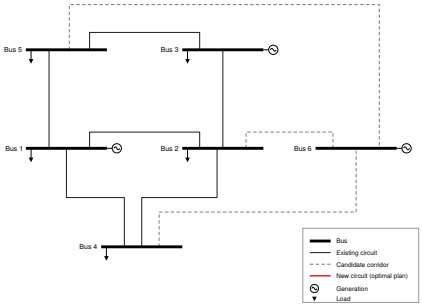
### *5.2. The Colombian 500 kV backbone*

#### 5.2.1. Nodal ranking

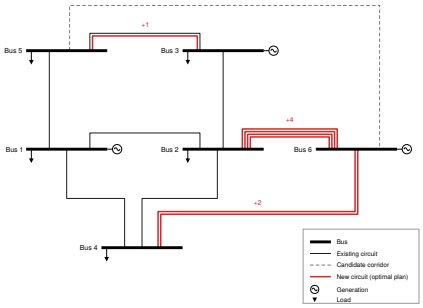
Table 6 reports the index for the ten highest-ranked nodes, and Figure 5 shows it for every node with demand. The ordering is dominated by the two grid-related terms, since irradiance varies by less than a factor of one and a half across the country while the supply deficit and the corridor loading vary by much more.

**Table 6.** Community Energy Index of the ten highest-ranked nodes

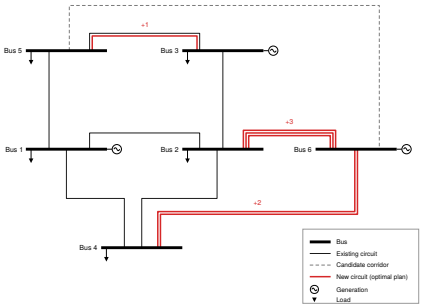
| Node | Substation | SPI | NSI | GRI | CEI |
|---|---|---|---|---|---|
| N23 | Norte | 0.742 | 1.000 | 0.463 | 0.736 |
| N13 | Nueva Esperanza | 0.742 | 0.429 | 1.000 | 0.725 |
| N18 | Sahagún | 0.887 | 1.000 | 0.191 | 0.712 |
| N3 | Bacatá | 0.742 | 0.529 | 0.753 | 0.682 |
| N14 | Ocaña | 0.774 | 0.837 | 0.394 | 0.679 |
| N20 | San Marcos | 0.758 | 0.639 | 0.511 | 0.648 |
| N10 | La Virginia | 0.758 | 0.205 | 0.898 | 0.634 |
| N11 | Medellín | 0.694 | 0.874 | 0.223 | 0.607 |
| N8 | Heliconia | 0.694 | 0.874 | 0.194 | 0.598 |
| N1 | Alférez | 0.758 | 0.503 | 0.435 | 0.585 |



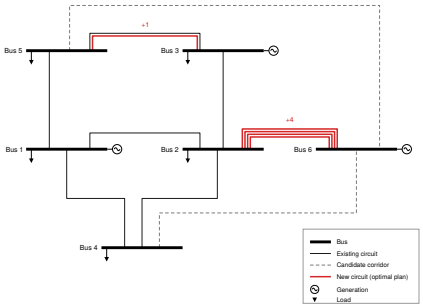

(a) Single-line diagram of the Garver test system



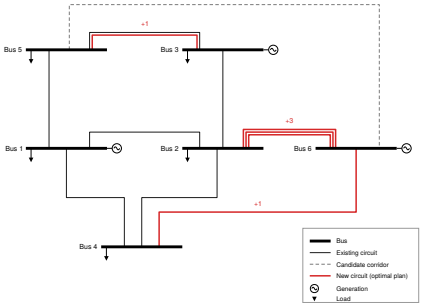

(b) $\alpha = 0$ %, 1000 kUSD



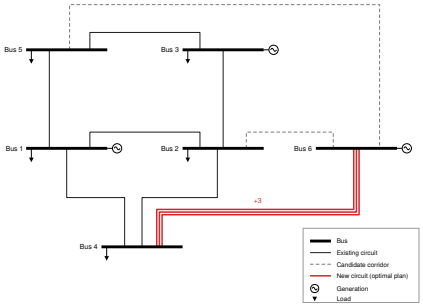

(c) $\alpha = 10$ %, 850 kUSD



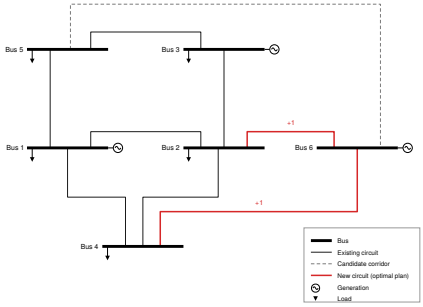

(d) $\alpha = 20$ %, 700 kUSD




(e) $\alpha = 30$ %, 700 kUSD

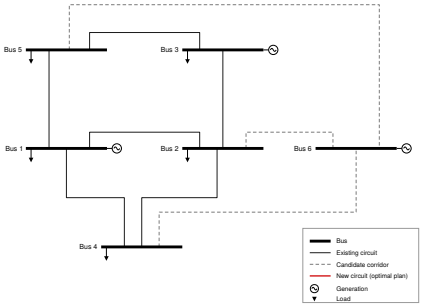



(f) $\alpha = 40$ %, 450 kUSD

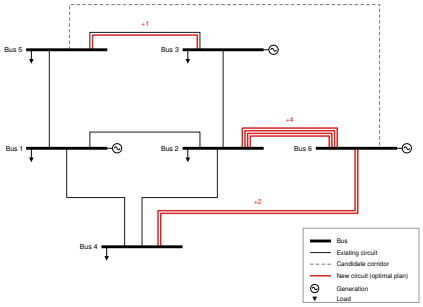



(g) $\alpha = 50$ %, 300 kUSD

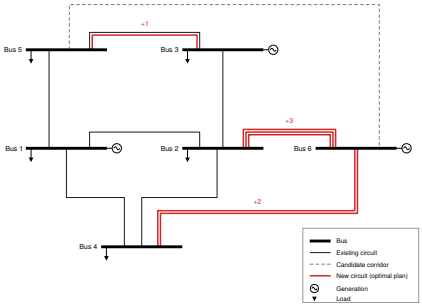
**Figure 3.** Optimal expansion plan of the Garver system for each behind-the-meter penetration level under the uniform allocation. Existing circuits in black, candidate corridors not built in grey dashed lines, and new circuits in red, with the number added to each corridor



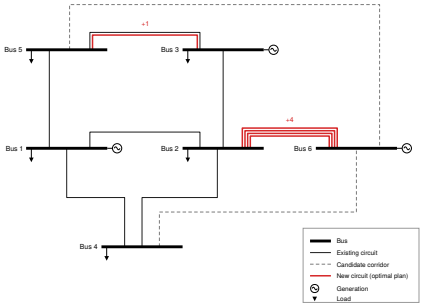

(a) Single-line diagram of the Garver test system



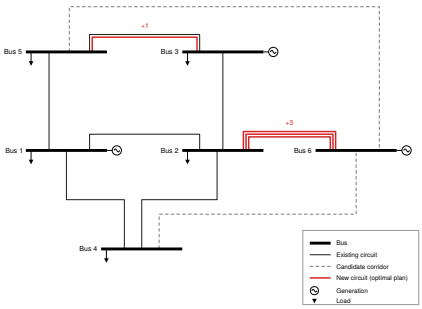

(b) $\alpha = 0$ %, 1000 kUSD



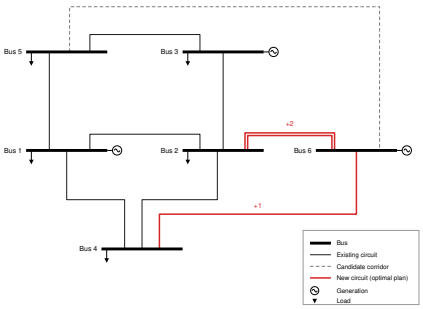

(c) $\alpha = 10$ %, 850 kUSD



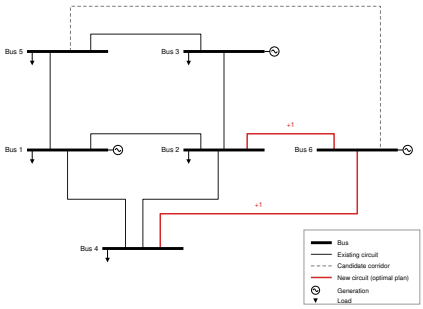

(d) $\alpha = 20$ %, 700 kUSD




(e) $\alpha = 30$ %, 550 kUSD




(f) $\alpha = 40$ %, 450 kUSD




(g) $\alpha = 50$ %, 300 kUSD

**Figure 4.** Optimal expansion plan of the Garver system for each behind-the-meter penetration level under the CEI allocation. Existing circuits in black, candidate corridors not built in grey dashed lines, and new circuits in red, with the number added to each corridor

The result is consistent with the physical reading of the system. The Bogotá-area nodes head the ranking together with Sahagún and Ocaña, all of which combine a large supply deficit with loaded corridors, while the generation hubs sit at the bottom: Porce III scores 0.289, Cerromatoso 0.306 and San Carlos 0.319. La Guajira is the clearest illustration of why the index cannot be built on the solar resource alone: Nueva Cuestecitas and Colectora have the highest irradiance in the country, and both reach the maximum SPI, yet neither is a priority because there is little or no local demand to reduce.

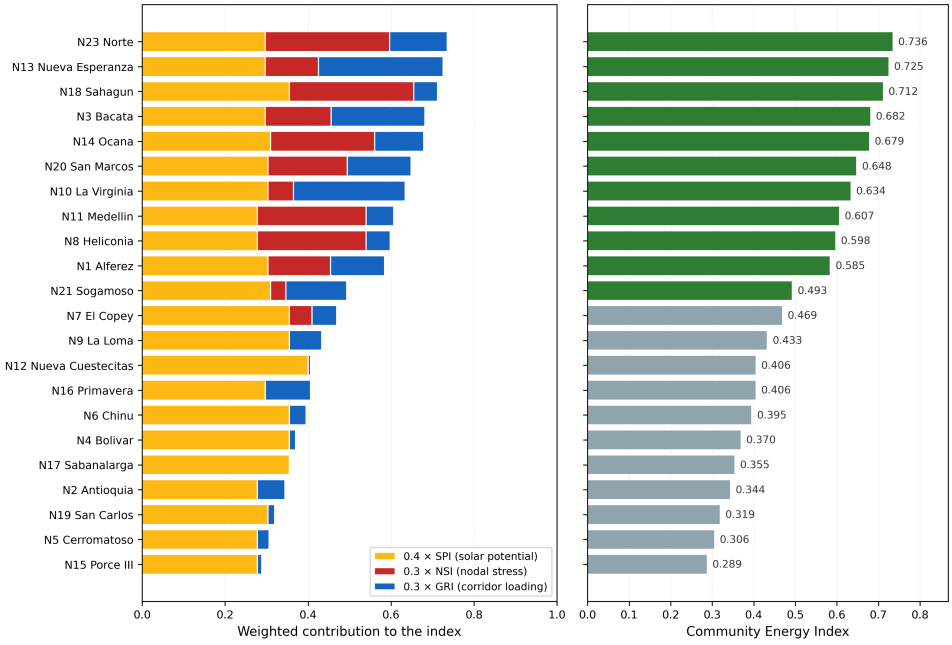



**Figure 5.** Community Energy Index by node. Left: decomposition into the SPI, NSI and GRI sub-indices. Right: composite index ordered from highest to lowest

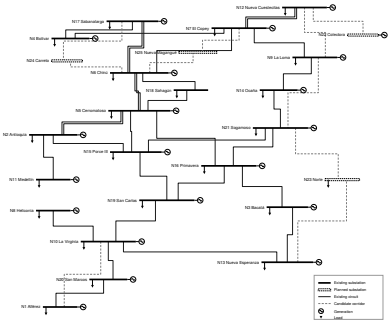
### 5.2.2. Base expansion plan

With no behind-the-meter resource, the high-electrification scenario requires five circuits for a total of 201.0 MUSD, listed in Table 7. Three of them reinforce existing corridors into the Bogotá area and the south-west, one reinforces the internal link of the Bogotá area, and one is a candidate project that connects the Norte substation, which is the entry point of the conditioned demand. Panel (b) of Figure 6 shows the location of these circuits on the network.

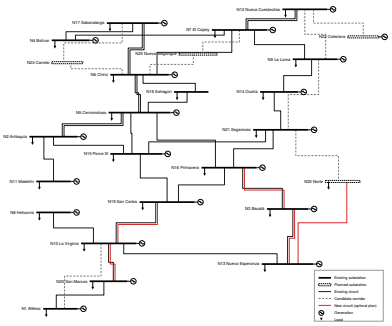
**Table 7.** Optimal expansion plan without behind-the-meter resource

| Corridor | Description | Type | Cost (MUSD) |
|---|---|---|---|
| N10–N19 | La Virginia–San Carlos | reinforcement | 60.25 |
| N3–N16 | Bacatá–Primavera | reinforcement | 55.92 |
| N10–N20 | La Virginia–San Marcos | reinforcement | 47.08 |
| N13–N23 | Nueva Esperanza–Norte | new corridor | 24.71 |
| N3–N13 | Bacatá–Nueva Esperanza | reinforcement | 13.04 |
| Total | 5 circuits | | 201.00 |

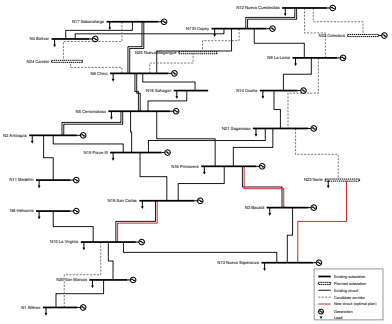
The plan follows the transfer pattern of the scenario. At the target year the Oriental area needs to import 3487 MW, and the corridors that carry that import, together with those that evacuate the Antioquia and San Carlos hubs towards the south-west, are the ones that saturate: La Virginia–San Carlos and Porce III–San Carlos operate at their rating and Bacatá–Primavera at 98 % of it. The plan is therefore not a general reinforcement of



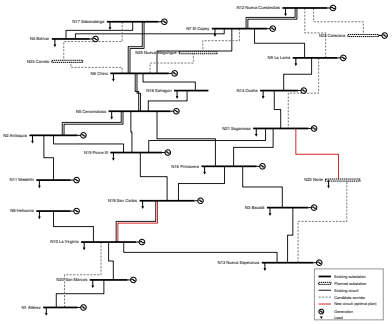
(a) Single-line diagram of the 500 kV backbone equivalent used in the case study



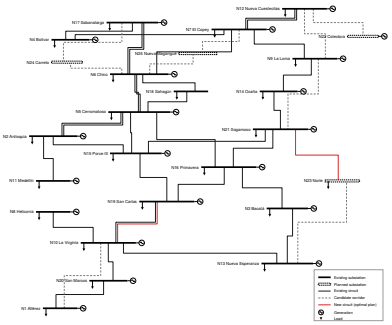
(b) $\alpha = 0$ %, 201.0 MUSD



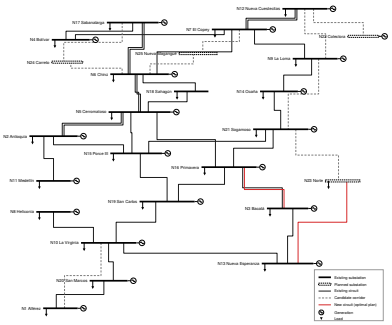
(c) $\alpha = 5$ %, 140.9 MUSD



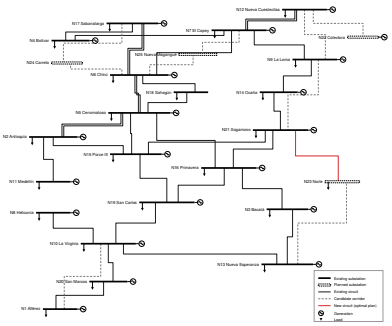
(d) $\alpha = 10$ %, 133.2 MUSD



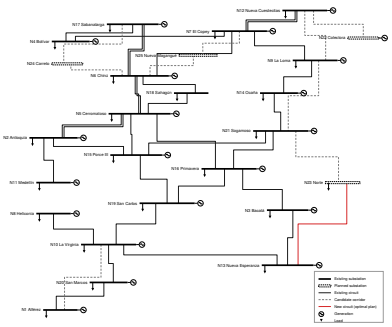
(e) $\alpha = 15$ %, 133.2 MUSD



(f) $\alpha = 20$ %, 80.6 MUSD

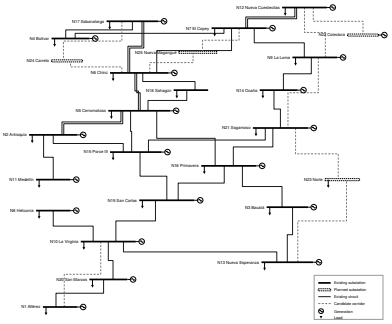


(g) $\alpha = 25$ %, 73.0 MUSD

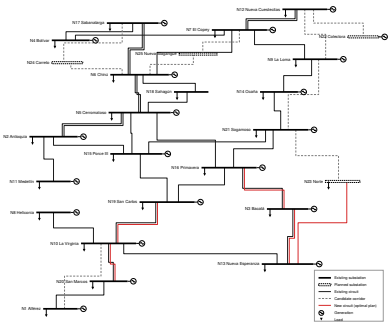


(h) $\alpha = 30$ %, 24.7 MUSD

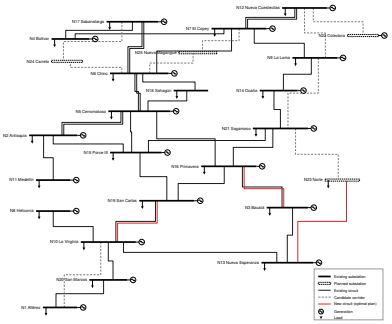
**Figure 6.** Optimal expansion plan of the 500 kV backbone for each behind-the-meter penetration level under the uniform allocation. Existing circuits in black, candidate corridors not built in grey dashed lines, and new circuits, including reinforcements of existing corridors, in red



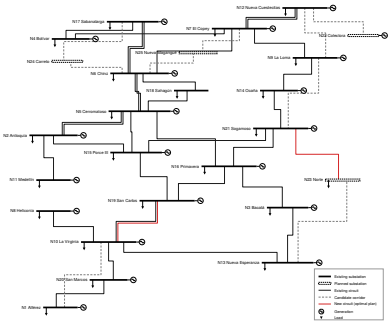
(a) Single-line diagram of the 500 kV backbone equivalent used in the case study



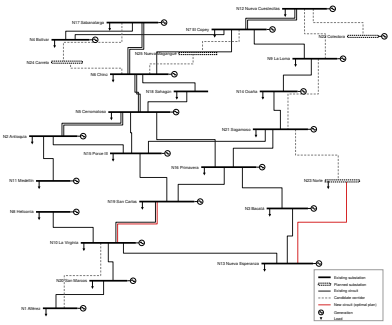
(b) $\alpha = 0$ %, 201.0 MUSD



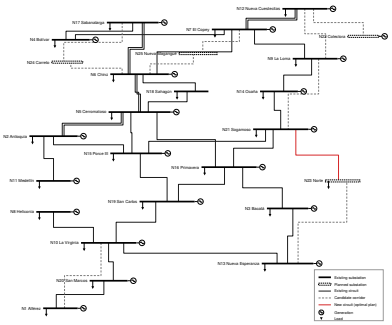
(c) $\alpha = 5$ %, 140.9 MUSD



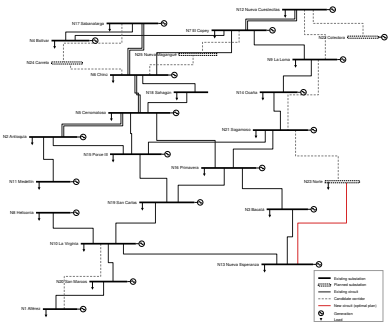
(d) $\alpha = 10$ %, 133.2 MUSD



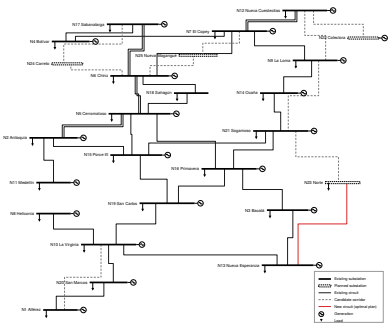
(e) $\alpha = 15$ %, 85.0 MUSD



(f) $\alpha = 20$ %, 73.0 MUSD



(g) $\alpha = 25$ %, 24.7 MUSD



(h) $\alpha = 30$ %, 24.7 MUSD

**Figure 7.** Optimal expansion plan of the 500 kV backbone for each behind-the-meter penetration level under the CEI allocation. Existing circuits in black, candidate corridors not built in grey dashed lines, and new circuits, including reinforcements of existing corridors, in red

the backbone but a targeted one, concentrated at two entry points, which is precisely the condition under which a local demand reduction can be worth a circuit.

#### 5.2.3. Effect of behind-the-meter penetration

Table 8 reports the optimal investment for penetrations between 0 and 30 % under both allocations.

**Table 8.** Optimal investment and savings by penetration level and allocation strategy

| $\alpha$ | $d^{\text{net}}$ (MW) | Uni. (MUSD) | CEI (MUSD) | Uni. saving (MUSD) | CEI saving (MUSD) |
|---|---|---|---|---|---|
| 0 % | 27925 | 201.0 | 201.0 | 0 | 0 |
| 5 % | 26529 | 140.9 | 140.9 | 60.1 | 60.1 |
| 10 % | 25133 | 133.2 | 133.2 | 67.8 | 67.8 |
| 15 % | 23736 | 133.2 | 85.0 | 67.8 | 116.0 |
| 20 % | 22340 | 80.6 | 73.0 | 120.4 | 128.0 |
| 25 % | 20944 | 73.0 | 24.7 | 128.0 | 176.3 |
| 30 % | 19548 | 24.7 | 24.7 | 176.3 | 176.3 |

The cost trajectory is monotone and stepwise, and Figure 6 shows the plan behind each step. A penetration of 5 % already removes two of the five circuits, those on Bacatá–Nueva Esperanza and La Virginia–San Marcos, and saves 60.1 MUSD (panel c): a 5 % reduction of net demand removes 30 % of the plan, six times its own magnitude, which is the clearest expression of the discrete nature of the mechanism. Between 10 and 15 % the uniform allocation stalls at 133.2 MUSD and the plan does not change at all (panels d and e), because the next circuit is not yet within reach; the CEI allocation, which concentrates the reduction on the Bogotá and Valle nodes, drops to 85.0 MUSD at 15 %.

Every plan in Figure 6 contains a circuit that connects the Norte substation, and this sets a floor that no penetration removes. That circuit is not there to relieve a transfer: Norte has no other link to the backbone and has to serve the conditioned demand of the area. Behind-the-meter resources reduce the amount of power the network transports, so they can remove reinforcements, but they cannot remove the connection of new load. At 30 % only the shortest connection, from Nueva Esperanza at 24.7 MUSD, remains (panel h), and the credit saturates at 176.3 MUSD, 88 % of the base plan, rather than at its full value.

Figure 7 shows the plans under the CEI allocation and Figure 8 compares both cost trajectories. The comparison is less regular than the intuition suggests. Targeting by CEI is never worse, and its advantage over the uniform allocation reaches 48.2 MUSD at 15 % and 48.3 MUSD at 25 %, close to the cost of a full circuit in each case, while at 20 % it is only 7.6 MUSD and at the remaining levels the two coincide. The reason is that both allocations sometimes fall on the same side of the threshold where a circuit leaves the plan. The value of siting is thus real but discrete: it materializes when the concentration of the reduction is enough to cross a threshold that the uniform allocation does not reach, and it vanishes when both cross it or neither does.

The figure also shows that, as in the Garver system, the plan is re-optimized rather than trimmed. At 10 % the connection of Norte moves from Nueva Esperanza to Sogamoso (panel d). The longer circuit costs 73.0 MUSD, but it brings power into the Bogotá area through a new path and relieves Bacatá–Primavera, so it does the job of two circuits that together cost 80.6 MUSD. At 20 % the reinforcement of La Virginia–San Carlos leaves the plan and the cheapest combination returns to the Nueva Esperanza connection with Bacatá–Primavera (panel f); at 25 % the Sogamoso connection alone is enough (panel g); and at 30 % only the short connection from Nueva Esperanza remains. Bacatá–Primavera thus leaves the plan at 10 % and comes back at 20 %, which a method that removes circuits from a fixed base plan could not reproduce.

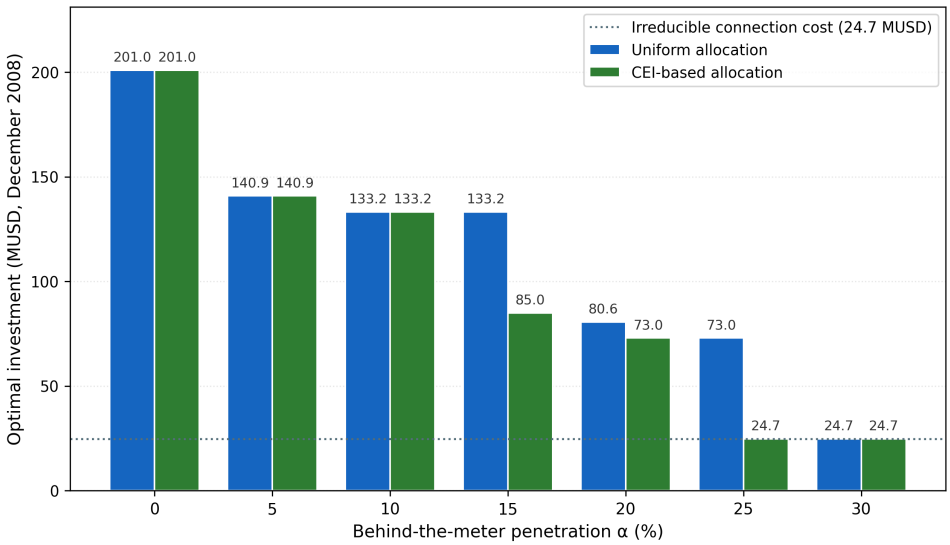



**Figure 8.** Optimal investment as a function of behind-the-meter penetration under the uniform and CEI-based allocations

5.2.4. Multi-period results and deferred credit

The multi-period model answers when each circuit is needed, and therefore separates what is avoided from what is only postponed. Table 9 reports the results for final penetrations of 10 and 20 % under the uniform allocation.

**Table 9.** Multi-period results over the 2026–2040 horizon, uniform allocation

| Scenario | Nominal (MUSD) | NPV (MUSD) | Deferred credit (MUSD) | Reduction (%) |
|---|---|---|---|---|
| Base, $\alpha = 0$ | 201.0 | 90.8 | – | – |
| $\alpha = 10$ % | 140.9 | 65.8 | 24.9 | 27.5 |
| $\alpha = 20$ % | 80.6 | 42.1 | 48.7 | 53.7 |

The schedule in Table 10 and Figure 9 shows where the credit comes from. In every scenario the connection of the Norte substation is built in 2027, which confirms that this element is driven by load connection and not by transfer. The reinforcement of La Virginia–San Carlos moves from 2033 to 2035 when penetration reaches 10 % and disappears at 20 %. Bacatá–Primavera moves from 2036 to 2038 and then to 2037 in the scenario where it is the only reinforcement left. The two circuits that the base plan builds in 2040 are never built under either penetration.

**Table 10.** Annual investment of the multi-period plan (MUSD). Only years with investment are shown

| Year | $\alpha = 0$ | $\alpha = 10$ % | $\alpha = 20$ % |
|---|---|---|---|
| 2027 | 24.7 | 24.7 | 24.7 |
| 2033 | 60.3 | – | – |
| 2035 | – | 60.3 | – |
| 2036 | 55.9 | – | – |
| 2037 | – | – | 55.9 |
| 2038 | – | 55.9 | – |
| 2040 | 60.1 | – | – |

The credit therefore has the two components it is supposed to have. Part of it is investment that leaves the plan altogether, and part is investment that is still made but one to two years later, which the discount rate turns into a present-value saving. The relation between penetration and credit remains monotone in present value, and the reduction is larger in relative terms than in nominal terms, 27.5 % and 53.7 % of the base present value against 30 % and 60 % of the nominal plan, because the circuits that survive are pushed towards the end of the horizon.

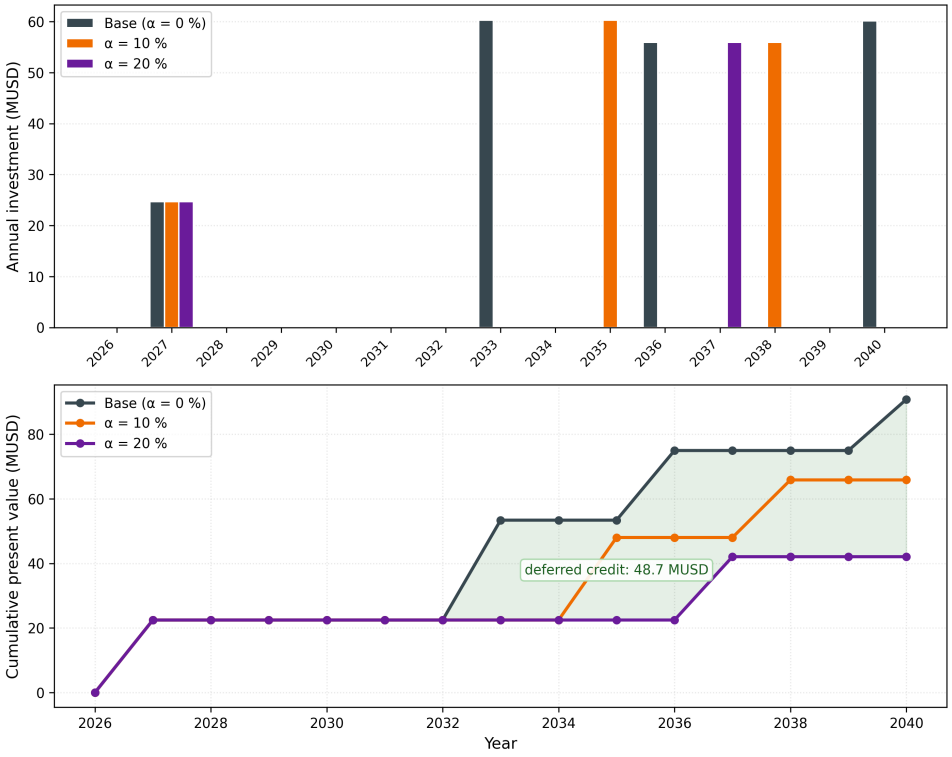



**Figure 9.** Multi-period plan for each final penetration level. Top: annual investment. Bottom: cumulative present value, where the shaded area is the deferred credit

### *5.3. Magnitude of the results and monetary base*

The figures reported above are expressed in December 2008 US dollars, which is the base of the reference construction units used to cost the circuits [40]. Two implications follow. First, the plan of 201 MUSD is not comparable to the budget of a project tendered today; an escalation to current dollars would raise it by roughly the accumulated inflation of the period. Second, and more useful for the purposes of this paper, the escalation is a single multiplicative factor applied to every corridor, because all of them are costed with the same unit prices. Indexing would therefore scale the base plan, the savings and the deferred credit by the same amount, and would leave unchanged the share of the plan that each penetration level removes, the comparison between the two allocation strategies and the shape of the credit curve. The results should be read as relative magnitudes, which is also how they are used: a penetration of 5 % removes 30 % of the plan, and a final penetration of 20 % removes slightly more than half of its present value.

### *5.4. Limitations*

The following limitations bound the interpretation of these results.

The models impose no security criterion. Circuits are limited by their thermal rating, so the reported plans are the smallest set of circuits that keeps the system feasible under normal operation. A study with an N-1 criterion would find a larger base plan, since the secure transfer limit of a corridor is well below its thermal rating, and would most likely find a larger credit as well, because relieving a corridor is worth more when its usable

capacity is smaller. The absolute figures are therefore conservative and should not be read as a substitute for a security-constrained plan.

Only the 500 kV level is represented. Reinforcements at lower voltages and transformation capacity, which in the Colombian system carry a significant part of the restriction in the Bogotá area, fall outside the model and are not credited to the behind-the-meter resource.

The nodal allocation of demand and generation is an approximation built from area projections and plant locations, as described in Section 4. It determines which corridors saturate and therefore which circuits the plan contains, so it is the input to which the results are most sensitive.

Generation is also dispatched without cost within its capacity and the penetration $\alpha$ is exogenous. The first means the optimizer chooses the dispatch that best relieves the network among all feasible ones, which is not necessarily the dispatch the market would produce. The second means the models measure what a given level of self-supply would be worth to the transmission system, not how much self-supply a community would find profitable to install. Making $\alpha$ the result of a community-level decision, in the spirit of the bi-level formulations used in transactive settings [5,24], is the natural extension of this work.

Two further choices condition the multi-period figures. The demand projection assumes, as stated in Section 4, that new large loads such as data centres connect behind the meter; if they connected directly to the backbone, they would add to $d_i$ and part of the credit would be absorbed by that additional demand. And the multi-period model was solved under the uniform allocation only. The static results suggest that a CEI-based allocation would remove the same circuits at lower aggregate penetration and therefore bring the credit forward, but extending the multi-period analysis to that allocation is left for future work.

## 6. Conclusions

This paper proposed a transmission expansion methodology in which behind-the-meter self-supply enters the planning problem as a nodal reduction of net demand, $d_i(1-\alpha_i)$, and a Community Energy Index ranks where that reduction is most valuable. Applied to the Garver system and to the Colombian 500 kV backbone, it leads to the following conclusions.

- Behind-the-meter self-supply works as a virtual transmission resource, and its value can be measured inside a standard expansion model. The methodology requires no change to the network formulation and applies to any system with significant penetration, so the planner can read the avoided and deferred investment directly from the optimal plans.
- The credit is discrete, not proportional. Small penetration levels remove whole circuits: in the Colombian case a 5 % reduction of net demand removes 30 % of the plan, while in other intervals additional penetration produces no saving until the next threshold is crossed.
- There is a floor that the resource cannot remove. Behind-the-meter supply reduces what the network transports, not the need to connect new load. The circuit that connects a new load centre remains in every plan, so the credit saturates below the full investment, at 88 % of the base plan in the Colombian case.
- Location matters, but in steps. Concentrating penetration on the nodes with the highest index is never worse than spreading it uniformly, and is worth close to a full circuit when it crosses a threshold the uniform allocation does not reach. The priority nodes are load centres with a supply deficit and loaded corridors, not the nodes with the best solar resource.
- The credit combines avoided and postponed investment. Over the planning horizon, part of the saving comes from circuits that are never built and part from circuits built

one to two years later; final penetrations of 10 % and 20 % reduce the present value of the Colombian plan by 27.5 % and 53.7 %.

- The plan is re-optimized, not trimmed. In both systems each penetration level produces its own topology, and some circuits leave the plan and return at a higher level. Estimates that remove circuits from a fixed base plan miss this effect.
- The results support targeting energy community incentives by location. In systems where new large loads, data centres included, connect behind the meter, self-supply adds to the credit instead of being offset, and an index such as the CEI offers a transparent criterion to direct public resources.

The figures of the Colombian case are expressed in December 2008 US dollars and should be read as relative magnitudes. The analysis imposes no security criterion, represents only the 500 kV level and treats penetration as exogenous, so the reported credit should be read as a conservative estimate. Natural extensions are an N-1 criterion, the sub-transmission and transformation levels, the multi-period evaluation of the CEI allocation, operating costs in the objective, and a bi-level formulation that couples expansion planning with the optimal sizing of behind-the-meter generation and storage.

## Appendix A. Nomenclature

*Acronyms*

| | |
|---|---|
| AGGE | Large-scale self-generation (autogeneración a gran escala) |
| AGPE | Small-scale self-generation (autogeneración a pequeña escala) |
| BTM | Behind the meter |
| CEI | Community Energy Index |
| DC | Direct current (linearized power flow) |
| DER | Distributed energy resources |
| DG | Distributed generation |
| FACTS | Flexible AC transmission systems |
| GHI | Global horizontal irradiance |
| GRI | Grid reinforcement index |
| kUSD, MUSD | Thousands and millions of US dollars |
| MILP | Mixed-integer linear programming |
| NPV | Net present value |
| NSI | Nodal stress index |
| PV | Photovoltaic |
| RES | Renewable energy sources |
| SPI | Solar potential index |
| TEP | Transmission expansion planning |

*Sets and indices*

| | |
|---|---|
| $\mathcal{I}$ | Set of nodes, indexed by $i$ |
| $\mathcal{J}$ | Set of corridors, indexed by $j$ |
| $\mathcal{K}$ | Set of circuit positions per corridor, indexed by $k$ |
| $\mathcal{T}$ | Set of periods, indexed by $t$ |
| $\Omega_i$ | Set of corridors incident to node $i$ |
| $a, b$ | Sending and receiving ends of corridor $j$ |

*Parameters*

| | |
|---|---|
| $B_j$ | Susceptance of corridor $j$ (p.u.) |
| $c_j$ | Cost of one circuit of corridor $j$ (MUSD) |
| $d_i$ | Gross demand at node $i$ (MW) |
| $F_j^{\max}$ | Thermal rating of one circuit of corridor $j$ (MW) |
| $G_i^{\max}$ | Generation capacity available at node $i$ (MW) |
| $GHI_i$ | Global horizontal irradiance at node $i$ (kWh/m$^2$ per day) |
| $M_j$ | Disjunctive constant of corridor $j$ (p.u.) |
| $n_j^0$ | Circuits of corridor $j$ already in service |
| $r$ | Discount rate |
| $S_{ij}$ | Node-corridor incidence |
| $T$ | Number of periods |
| $X_j$ | Reactance of corridor $j$ (p.u.) |
| $\alpha_i$ | Behind-the-meter penetration at node $i$ |
| $\alpha^{\max}$ | Maximum penetration allowed at a node |
| $\bar{\alpha}$ | Aggregate penetration level |
| $\Gamma$ | Investment budget (MUSD) |
| $\Delta\theta^{\max}$ | Bound on the angular difference between nodes (rad) |
| $\hat{f}_j$ | Flow of corridor $j$ in the base case (MW) |
| $\eta_N, \eta_G$ | Normalization factors of the NSI and GRI sub-indices |
| $d_i^{\text{net}}$ | Net demand at node $i$ (MW) |
| $\alpha_{i,t}$ | Behind-the-meter penetration at node $i$ in period $t$ |
| $\overline{CEI}$ | Mean index over the nodes with demand |
| $SPI_i, NSI_i, GRI_i, CEI_i$ | Sub-indices and composite index of node $i$ |

*Variables*

| | |
|---|---|
| $f_{j,k}$ | Power flow through circuit $k$ of corridor $j$ (p.u.) |
| $g_i$ | Power injected at node $i$ (p.u.) |
| $x_{j,k,t}$ | 1 if circuit $k$ of corridor $j$ is available in period $t$ |
| $y_{j,k}$ | 1 if circuit $k$ of corridor $j$ is in service (static model) |
| $y_{j,k,t}$ | 1 if circuit $k$ of corridor $j$ is built in period $t$ |
| $\theta_i$ | Voltage angle at node $i$ (rad) |

*Metrics*

| | |
|---|---|
| $C^{\text{av}}(\alpha)$ | Avoided investment (MUSD) |
| $C^{\text{def}}(\alpha)$ | Deferred investment credit in present value (MUSD) |
| $I^{\text{base}}$ | Cost of the optimal plan without behind-the-meter resource (MUSD) |
| $I_t$ | Investment committed in period $t$ (MUSD) |
| $I(\alpha)$ | Cost of the optimal plan under penetration $\alpha$ (MUSD) |

## Appendix B. Test system data

### *Appendix B.1. Garver system*

Table A1 lists the corridor data of the six-bus system. Demand and generation capacity are given in Table 2 of the main text. Corridors with no assigned cost are not candidates in this variant of the problem.

**Table A1.** Corridor data of the Garver system

| Corridor | Existing circuits | $B_j$ (p.u.) | $F_j^{\max}$ (MW) | $c_j$ (kUSD) |
|---|---|---|---|---|
| 1–2 | 1 | 2.50 | 100 | – |
| 1–3 | 0 | – | 100 | – |
| 1–4 | 1 | 1.67 | 80 | – |
| 1–5 | 1 | 5.00 | 100 | – |
| 1–6 | 0 | – | 70 | – |
| 2–3 | 1 | 5.00 | 100 | – |
| 2–4 | 1 | 2.50 | 100 | – |
| 2–5 | 0 | – | 100 | – |
| 2–6 | 0 | 3.33 | 100 | 150 |
| 3–4 | 0 | – | 82 | – |
| 3–5 | 1 | 5.00 | 100 | 100 |
| 3–6 | 0 | – | 100 | – |
| 4–5 | 0 | – | 75 | – |
| 4–6 | 0 | 3.33 | 100 | 150 |
| 5–6 | 0 | 1.64 | 78 | 350 |

*Appendix B.2. Colombian 500 kV backbone*

Table A2 lists the 37 corridors of the equivalent network, 27 of them already in service with 31 circuits and 10 associated with planned projects. Reactances are expressed on a 100 MVA base, ratings correspond to the average of the circuits of the corridor, and costs are per circuit in December 2008 US dollars. Nodal demand and generation are given in Table 3 of the main text.

**Table A2.** Corridor data of the 500 kV backbone

| Corridor | Substations | $n_j^0$ | Length (km) | $X_j$ (p.u.) | $F_j^{\max}$ (MW) | $c_j$ (MUSD) |
|---|---|---|---|---|---|---|
| N1–N20 | Alférez–San Marcos | 1 | 36 | 0.0046 | 2078 | 10.22 |
| N2–N5 | Antioquia–Cerromatoso | 2 | 113 | 0.0144 | 2078 | 31.97 |
| N2–N11 | Antioquia–Medellín | 1 | 110 | 0.0141 | 2078 | 31.12 |
| N2–N15 | Antioquia–Porce III | 1 | 113 | 0.0145 | 2078 | 32.03 |
| N3–N13 | Bacatá–Nueva Esperanza | 1 | 46 | 0.0064 | 2165 | 13.04 |
| N3–N16 | Bacatá–Primavera | 1 | 197 | 0.0261 | 1650 | 55.92 |
| N4–N7 | Bolívar–El Copey | 1 | 167 | 0.0223 | 1650 | 43.33 |
| N4–N17 | Bolívar–Sabanalarga | 1 | 67 | 0.0085 | 2078 | 17.30 |
| N5–N6 | Cerromatoso–Chinú | 2 | 134 | 0.0171 | 2122 | 38.13 |
| N5–N15 | Cerromatoso–Porce III | 1 | 142 | 0.0187 | 1650 | 40.23 |
| N5–N16 | Cerromatoso–Primavera | 1 | 245 | 0.0316 | 2165 | 69.64 |
| N5–N18 | Cerromatoso–Sahagún | 1 | 79 | 0.0110 | 2078 | 22.46 |
| N6–N7 | Chinú–El Copey | 1 | 233 | 0.0290 | 2078 | 60.46 |
| N6–N17 | Chinú–Sabanalarga | 2 | 184 | 0.0243 | 2111 | 47.80 |
| N6–N18 | Chinú–Sahagún | 1 | 53 | 0.0074 | 2078 | 13.78 |
| N7–N9 | El Copey–La Loma | 1 | 73 | 0.0104 | 1650 | 18.98 |
| N7–N12 | El Copey–Nueva Cuestecitas | 2 | 218 | 0.0270 | 2078 | 56.74 |
| N8–N10 | Heliconia–La Virginia | 1 | 178 | 0.0231 | 2078 | 50.69 |
| N9–N14 | La Loma–Ocaña | 1 | 168 | 0.0239 | 1650 | 47.71 |
| N10–N13 | La Virginia–Nueva Esperanza | 1 | 238 | 0.0305 | 2078 | 67.47 |
| N10–N19 | La Virginia–San Carlos | 1 | 212 | 0.0268 | 1732 | 60.25 |
| N10–N20 | La Virginia–San Marcos | 1 | 166 | 0.0210 | 1732 | 47.08 |
| N14–N21 | Ocaña–Sogamoso | 1 | 164 | 0.0216 | 1650 | 46.49 |
| N15–N19 | Porce III–San Carlos | 1 | 106 | 0.0140 | 1650 | 30.24 |
| N15–N21 | Porce III–Sogamoso | 1 | 236 | 0.0299 | 2078 | 66.94 |
| N16–N19 | Primavera–San Carlos | 1 | 83 | 0.0107 | 2165 | 23.47 |
| N16–N21 | Primavera–Sogamoso | 1 | 136 | 0.0180 | 1650 | 38.53 |
| N1–N10 | Alférez–La Virginia | 0 | 183 | 0.0237 | 2078 | 51.97 |
| N6–N24 | Chinú–Carreto | 0 | 180 | 0.0233 | 2078 | 46.79 |
| N6–N25 | Chinú–Nueva Magangué | 0 | 95 | 0.0123 | 2078 | 24.69 |
| N7–N25 | El Copey–Nueva Magangué | 0 | 140 | 0.0181 | 2078 | 36.39 |
| N9–N12 | La Loma–Nueva Cuestecitas | 0 | 220 | 0.0285 | 2078 | 57.19 |
| N9–N21 | La Loma–Sogamoso | 0 | 275 | 0.0356 | 2078 | 78.10 |
| N12–N22 | Nueva Cuestecitas–Colectora | 0 | 220 | 0.0285 | 2078 | 57.19 |
| N13–N23 | Nueva Esperanza–Norte | 0 | 87 | 0.0113 | 2078 | 24.71 |
| N17–N24 | Sabanalarga–Carreto | 0 | 110 | 0.0142 | 2078 | 28.59 |
| N21–N23 | Sogamoso–Norte | 0 | 257 | 0.0332 | 2078 | 72.99 |

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